

DreamerVLA: Integrating Vision-Language-Action Models with Latent World Models

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Abstract: Embodied intelligence research mainly follows two directions: Vision-Language-Action (VLA) models and world model-based reinforcement learning methods. VLA models benefit from strong semantic understanding provided by large vision-language models, but they often lack the ability to model environment dynamics and perform long-horizon reasoning. In contrast, world model approaches such as Dreamer can perform long-horizon planning through latent dynamics models, but they typically have weaker perception and limited semantic understanding. In this work, we explore DreamerVLA, a framework that combines the strengths of both approaches by embedding multimodal observations with a frozen VLA encoder, learning compact latent dynamics, and optimizing an actor-critic policy through imagined rollouts. This framework aims to integrate semantic perception with model-based planning for embodied decision making.

Keywords: Vision-Language-Action Models, World Models, Robot Learning

1 Introduction

In embodied intelligence, an agent is required to predict the next action chunk given visual observations, language instructions, and proprioceptive signals. Existing methods can generally be categorized into two mainstream paradigms. The first is the Vision-Language-Action (VLA) paradigm, where the policy conditions on images, language instructions, and robot states to autoregressively generate an action chunk. The second is the world model paradigm, represented by Dreamer, which learns latent dynamics to predict future world states conditioned on actions, and employs an actor-critic framework to evaluate candidate actions and select the final action chunk.

Both paradigms, however, suffer from their own limitations. Although VLA models possess strong semantic understanding and can generate the required action chunk conditioned on visual observations and language instructions, they generally lack an explicit mechanism for modeling the long-term consequences of actions. In particular, VLA models struggle to reason about how different action magnitudes or variations may lead to different future outcomes. While this limitation may not cause severe errors in single-step prediction, it implies that the current action is generated without sufficiently considering subsequent action choices and their long-term effects, resulting in weak long-horizon decision-making ability. In contrast, world models explicitly model environment dynamics and therefore naturally support long-horizon reasoning. However, they are typically weaker in semantic understanding, making it difficult to control robot behavior through high-level language instructions. Moreover, pure world model approaches are challenging to scale in real-world settings, since real-world data are expensive to collect, and the mismatch between the learned world model and the real environment can accumulate over multiple prediction steps, leading to a significant gap between simulated performance and real-world deployment.

To combine the semantic generalization ability of VLA with the dynamics-aware optimization capability of world models, we propose DreamerVLA, a unified framework that trains VLA under the Dreamer paradigm. Our key observation is that the actor in Dreamer is functionally similar to a

VLA policy, which makes it possible to replace the conventional actor with a VLA-based policy and incorporate VLA into the Dreamer training framework. Specifically, during training, we first use a pretrained VLA encoder to encode multimodal inputs, including text, images, and proprioceptive states, into an embedding vector that represents the current environment. This representation is then processed by a bottleneck module, which can be implemented using either TokenLearner or a simple MLP, in order to compress and refine the state representation. The resulting latent state is fed into a world model to predict the next environment state, while an actor-critic module is trained to evaluate the quality of state-action pairs and guide policy optimization. During inference, the agent can leverage the world model together with the actor-critic module to perform imagination-based decision-making, thereby enabling long-horizon planning and reasoning.

2 Related Work

Vision-Language-Action Models

World Model for Embodied Control

Reinforcement Learning for Vision-Language-Action Models Recent work has begun to move beyond pure imitation learning and study reinforcement learning as a post-training mechanism for Vision-Language-Action (VLA) models. iRe-VLA shows that directly applying online RL to large VLA policies is often unstable and computationally demanding, and therefore alternates between RL-based exploration and supervised learning to absorb successful trajectories more stably. RIPT-VLA instead advocates a simpler critic-free interactive post-training paradigm that uses only sparse binary success rewards, combining dynamic rollout sampling with leave-one-out advantage estimation for stable policy updates. To improve long-horizon credit assignment, TGRPO extends group-relative optimization with both step-level and trajectory-level relative advantages, relying on dense multi-stage rewards to provide finer-grained supervision. From a systems perspective, VLA-RL formulates auto-regressive VLA control as trajectory-level online RL and introduces a robotic process reward model, together with implementation strategies such as critic warmup and scalable batch decoding, to make large-scale online optimization practical. Complementary to direct environment interaction, VLA-RFT leverages a data-driven world model as a controllable simulator and fine-tunes VLA policies using verified dense trajectory-level rewards computed from imagined visual rollouts. Overall, these studies suggest three emerging directions for RL-enhanced VLA training: alternating RL with imitation-style updates, direct online post-training under sparse or shaped rewards, and world-model-based reinforcement fine-tuning in learned simulators.

3 Method

3.1 Overview

We propose **DreamerVLA**, a Dreamer-style training framework that combines the semantic prior of a pretrained Vision-Language-Action (VLA) model with the long-horizon optimization ability of latent world models. The current DreamerVLA pipeline uses a pi0-style action-query VLA head to produce action-conditioned hidden observations, trains a DreamerV3 recurrent state-space model (RSSM) over these hidden observations, and then optimizes an actor-critic policy through imagined latent rollouts.

At each time step t , the robot observes two RGB views, a language instruction, and proprioceptive state:

$$o_t = \{I_t^{\text{third}}, I_t^{\text{wrist}}, x_t^{\text{prop}}, l\}.$$

The VLA model first converts this multimodal observation into an action-query hidden sequence

$$u_t = f_{\text{VLA}}^{\text{query}}(o_t) \in \mathbb{R}^{K \times 1024},$$

where K is the action horizon. We flatten this sequence into the world-model observation

$$e_t = \text{vec}(u_t) \in \mathbb{R}^{5120},$$

using $K = 5$ in our LIBERO configuration. These hidden observations are precomputed and stored in
 sidecar HDF5 files aligned with the original demonstrations, so world-model and actor-critic training
 do not backpropagate through the large VLA backbone.

DreamerVLA contains three trainable components:

1. an **action-hidden DreamerV3 world model** that uses pi0 action-query hidden states as ob-
 servations for a discrete RSSM while reconstructing pixels and action-hidden observations;
2. a **pi0 action-hidden actor** that maps imagined RSSM features back to action-query hidden
 states and reuses the pretrained pi0 output projection to produce action chunks;
3. a **two-hot critic** trained from DreamerV3-style imagined λ -returns.

This design keeps the VLA backbone fixed as a semantic action-conditioned encoder, learns compact
 task dynamics in latent space, and optimizes the action head through imagined rollouts without
 additional environment interaction.

3.2 Overall Training Pipeline

DreamerVLA is trained in four stages. First, we prepare LIBERO demonstrations by removing no-op
 transitions and storing synchronized third-person RGB, wrist RGB, robot states, actions, rewards, and
 terminal labels. The action horizon is set to $K = 5$, and each action is represented as a 7-dimensional
 continuous control vector.

Second, we warm-start the VLA policy with supervised action prediction. Starting from the pre-
 trained RynnVLA/Chameleon backbone, we fine-tune a pi0-style action-query head on LIBERO
 demonstrations. Instead of predicting actions directly from all language-image tokens, the head
 appends one learned action query for each future action step and regresses an action chunk from
 the resulting query hidden states. This stage provides the supervised VLA checkpoint used by all
 downstream DreamerVLA components.

Third, we freeze the VLA backbone and precompute action-conditioned hidden observations for
 every frame. The resulting sidecar files store e_t together with metadata for the VLA checkpoint,
 action horizon, action-head type, and hidden-source type. The dataset loader checks this metadata at
 startup, preventing stale hidden sidecars from being mixed with incompatible checkpoints.

Fourth, we train the action-hidden DreamerV3 world model and then optimize the actor and critic
 with imagined rollouts. World-model pretraining can be run as a standalone stage, and the resulting
 checkpoint can be loaded into joint DreamerVLA training. In the actor-critic stage, posterior latent
 states inferred from replay windows initialize imagined trajectories; the actor produces action chunks
 from imagined hidden states, and the first action in each chunk advances the RSSM rollout.

3.3 Frozen VLA Action-Query Features

We use the pretrained RynnVLA/Chameleon model as a frozen multimodal encoder. For each frame,
 the third-person view, wrist-camera view, task text, and optional proprioceptive state are serialized
 into the prompt format used by the VLA model. The Chameleon image tokenizer and text tokenizer
 convert the input into a unified token sequence, which is processed by the frozen Transformer
 backbone.

The pi0-style action-query head appends K learned query tokens, one per future action step. Context
 tokens condition these queries through attention, and the head exposes the query outputs as compact
 action-conditioned hidden states:

$$u_t = [u_t^1, \dots, u_t^K], \quad u_t^k \in \mathbb{R}^{1024}.$$

The world model uses their flattened representation $e_t = \text{vec}(u_t)$ as its observation. Compared
 with storing a full VLA token sequence, this representation is smaller and directly aligned with the
 downstream action head. During Dreamer training, the VLA backbone remains frozen and is not
 included in the optimization graph.

127 3.4 Action-Hidden DreamerV3 World Model

128 The world model follows the DreamerV3 recurrent state-space model, but replaces the standard
 129 pixel encoder with a lightweight encoder over precomputed action-hidden observations. Given
 130 action-hidden observations $\{e_t\}_{t=1}^T$ and actions $\{a_t\}_{t=1}^T$, the encoder maps each e_t into the RSSM
 131 observation space. The RSSM state is

$$s_t = (h_t, z_t),$$

132 where $h_t \in \mathbb{R}^{8192}$ is a deterministic recurrent state and z_t is a discrete stochastic state with 32
 133 variables and 64 categories per variable. The flattened latent feature is

$$\phi_t = [h_t; \text{vec}(z_t)] \in \mathbb{R}^{10240}.$$

134 **RSSM transition and posterior.** The deterministic transition is a DreamerV3 block-GRU update
 135 conditioned on the previous deterministic state, stochastic state, and action:

$$h_t = g_\theta(h_{t-1}, z_{t-1}, a_t).$$

136 The model predicts a prior distribution $p_\theta(z_t | h_t)$ and an observation-conditioned posterior

$$q_\theta(z_t | h_t, e_t).$$

137 We use the DreamerV3 unimix categorical parameterization and split the KL regularizer into dynamics
 138 and representation terms:

$$\mathcal{L}_{\text{dyn}} = \text{KL}(\text{sg}(q_\theta(z_t | h_t, e_t)) \| p_\theta(z_t | h_t)),$$

139

$$\mathcal{L}_{\text{rep}} = \text{KL}(q_\theta(z_t | h_t, e_t) \| \text{sg}(p_\theta(z_t | h_t))).$$

140 **Multi-target reconstruction.** The RSSM is trained with both visual and action-hidden recon-
 141 struction losses. A pixel decoder reconstructs the concatenated third-person and wrist images from
 142 ϕ_t :

$$\hat{I}_t = d_{\text{img}}(\phi_t), \quad \mathcal{L}_{\text{img}} = \|\hat{I}_t - I_t\|_2^2.$$

143 A hidden decoder predicts the flattened pi0 action-query hidden observation:

$$\hat{e}_t = d_{\text{hid}}(\phi_t), \quad \mathcal{L}_{\text{hid}} = \|\hat{e}_t - e_t\|_2^2.$$

144 This hidden reconstruction is the bridge between the RSSM and the pi0 action-hidden actor: during
 145 imagination, the actor consumes $\hat{e}_t$ rather than raw pixels or full VLA token states.

146 **Reward and continuation.** The world model predicts sparse task reward and continuation from
 147 the latent feature:

$$\hat{r}_t = d_r(\phi_t), \quad \hat{c}_t = d_c(\phi_t).$$

148 For LIBERO-style sparse success rewards, the implementation uses a binary reward head. The
 149 continuation head is trained with a binary cross-entropy objective over non-terminal targets.

150 The full world-model objective is

$$\mathcal{L}_{\text{WM}} = \lambda_{\text{img}} \mathcal{L}_{\text{img}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{rep}} \mathcal{L}_{\text{rep}} + \lambda_{\text{hid}} \mathcal{L}_{\text{hid}} + \lambda_r \mathcal{L}_r + \lambda_c \mathcal{L}_c.$$

151 3.5 Pi0 Action-Hidden Actor

152 Instead of using a small MLP policy, DreamerVLA reuses the action prior learned by the supervised
 153 pi0-style VLA head. The world model first decodes an imagined latent feature into a reconstructed
 154 action-hidden vector,

$$\hat{e}_t = d_{\text{hid}}(\phi_t) \in \mathbb{R}^{5120},$$

155 which is reshaped into

$$\hat{u}_t \in \mathbb{R}^{K \times 1024}.$$

A lightweight adapter refines this sequence, and the pretrained pi0 output projection maps each action-query hidden state to a continuous action. The actor therefore outputs an action chunk

$$\mu_{t:t+K-1} = \pi_\psi(\hat{u}_t) \in \mathbb{R}^{K \times A}.$$

In our LIBERO configuration, $K = 5$ and $A = 7$. During imagination, the first action in the chunk is used as the action mean,

$$\mu_t = \mu_{t:t+K-1}[0],$$

and a learned diagonal log-standard deviation defines a Gaussian policy:

$$a_t \sim \mathcal{N}(\mu_t, \text{diag}(\sigma^2)).$$

The actor can freeze or fine-tune the pi0 output projection. In both cases, the policy is initialized from the supervised VLA checkpoint and further optimized by DreamerV3 actor-critic updates.

3.6 Imagination-Based Actor-Critic Learning

Actor-critic training starts from posterior latent states inferred from replay sequences. From each start state, the world model generates an imagined trajectory of horizon H :

$$a_t \sim \pi_\psi(\cdot \mid \phi_t), \quad s_{t+1} \sim p_\theta(\cdot \mid s_t, a_t), \quad \hat{r}_{t+1} = d_r(\phi_{t+1}), \quad \hat{c}_{t+1} = d_c(\phi_{t+1}).$$

Gradients are stopped through imagined world-model states, matching the standard DreamerV3 actor update.

The critic receives the RSSM feature ϕ_t and predicts a categorical distribution over symlog value bins. Its scalar value is the symexp-transformed expectation:

$$V_\omega(\phi_t) = \text{symexp}(\mathbb{E}_{b \sim p_\omega(\cdot \mid \phi_t)}[b]).$$

A Polyak-averaged target critic provides bootstrap values. We compute continuation-weighted λ -returns:

$$R_t^\lambda = \hat{r}_{t+1} + \gamma \hat{c}_{t+1} ((1 - \lambda)V_{\bar{\omega}}(\phi_{t+1}) + \lambda R_{t+1}^\lambda).$$

The actor is trained with a likelihood-ratio objective using the imagined advantage:

$$\mathcal{L}_\pi = -\mathbb{E} \left[w_t \left(\log \pi_\psi(a_t \mid \phi_t) \text{sg} \left(\frac{R_t^\lambda - V_\omega(\phi_t)}{S} \right) + \eta \mathcal{H}[\pi_\psi(\cdot \mid \phi_t)] \right) \right],$$

where w_t is the cumulative continuation discount and $S = \max(1, P_{95}^{\text{ema}} - P_5^{\text{ema}})$ is the running percentile return scale. The critic is trained by maximizing the log probability of the two-hot symlog target:

$$\mathcal{L}_V = -\mathbb{E} [\log p_\omega(\text{twohot}(\text{sg}(R_t^\lambda)) \mid \phi_t)].$$

3.7 Training Strategy

Training alternates between two phases on the same replay windows. First, the world model is updated using real image sequences, precomputed action-hidden observations, actions, rewards, and terminal labels. Second, the actor and critic are updated from imagined rollouts produced by the learned RSSM. Only the world model, pi0 action-hidden actor, and critic are optimized; the VLA backbone used to generate action-query hidden observations remains frozen throughout.

4 Experiments

4.1 Experimental Setup

Baselines

Table 1: Comparison with state-of-the-art methods on LIBERO.

Method	Spatial	Object	Goal	Long	Avg.
GR-1	–	–	–	–	–
RT-1	–	–	–	–	–
Diffusion Policy	–	–	–	–	–
Octo	–	–	–	–	–
OpenVLA	–	–	–	–	–
RynnVLA	–	–	–	–	–
DreamerVLA (Ours)	–	–	–	–	–

Table 2: Ablation study on LIBERO. Each row adds one component to the baseline.

Variant	WM	PPO	Adv.	KL	Spatial	Goal	Long	Avg.
Baseline (VLA only)					–	–	–	–
+ PPO		✓			–	–	–	–
+ PPO + WM	✓	✓			–	–	–	–
+ PPO + WM + Adv.	✓	✓	✓		–	–	–	–
Full model	✓	✓	✓	✓	–	–	–	–

4.2 Main Results

4.3 Ablation Study

4.4 Generalization and Robustness

5 Conclusion

We presented DreamerVLA, a framework that combines the semantic representation ability of pretrained Vision-Language-Action models with the long-horizon reasoning ability of latent world models. By keeping the VLA encoder frozen and training a compact latent dynamics model together with an imagination-based actor-critic module, DreamerVLA separates language-grounded perception from task-specific predictive control. This design avoids pixel-level reconstruction while enabling policy optimization through imagined latent rollouts. Future work will further evaluate the framework on broader real-world manipulation tasks and study how world-model uncertainty can be used to improve safe deployment.